Group Number:
Student Names:
Marks: 4 points

COMP 251: Data Structures & Algorithms
Group Assignment 9: Search Trees

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************* INSTRUCTIONS *****************

- **Box the final tree** in both questions so that it is clearly distinguishable from intermediate steps.
- Ensure that all tree diagrams are **neat and easy to read**. You may draw them by hand (scanned) or using software.
- Follow a logical and step-by-step approach; unsupported final answers without intermediate steps may receive little or no credit.
- All submitted work must be your own (or your group's) original work. Academic integrity policies apply.

Question 1: Binary Search Trees

Insert into an empty binary search tree, entries with keys 30, 40, 24, 58, 48, 26, 11, 13 (in this order). Draw the tree after each insertion.

Question 2: Insertion in AVL Trees

Draw the AVL tree resulting from the insertion of an entry with key **52** into the tree shown in the Figure 1, given on the next page.

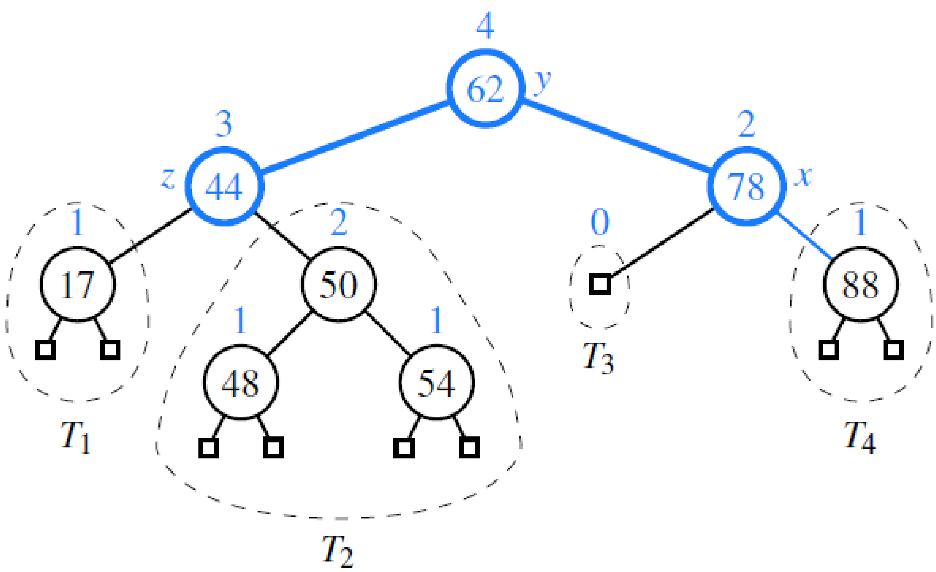

Figure 1: